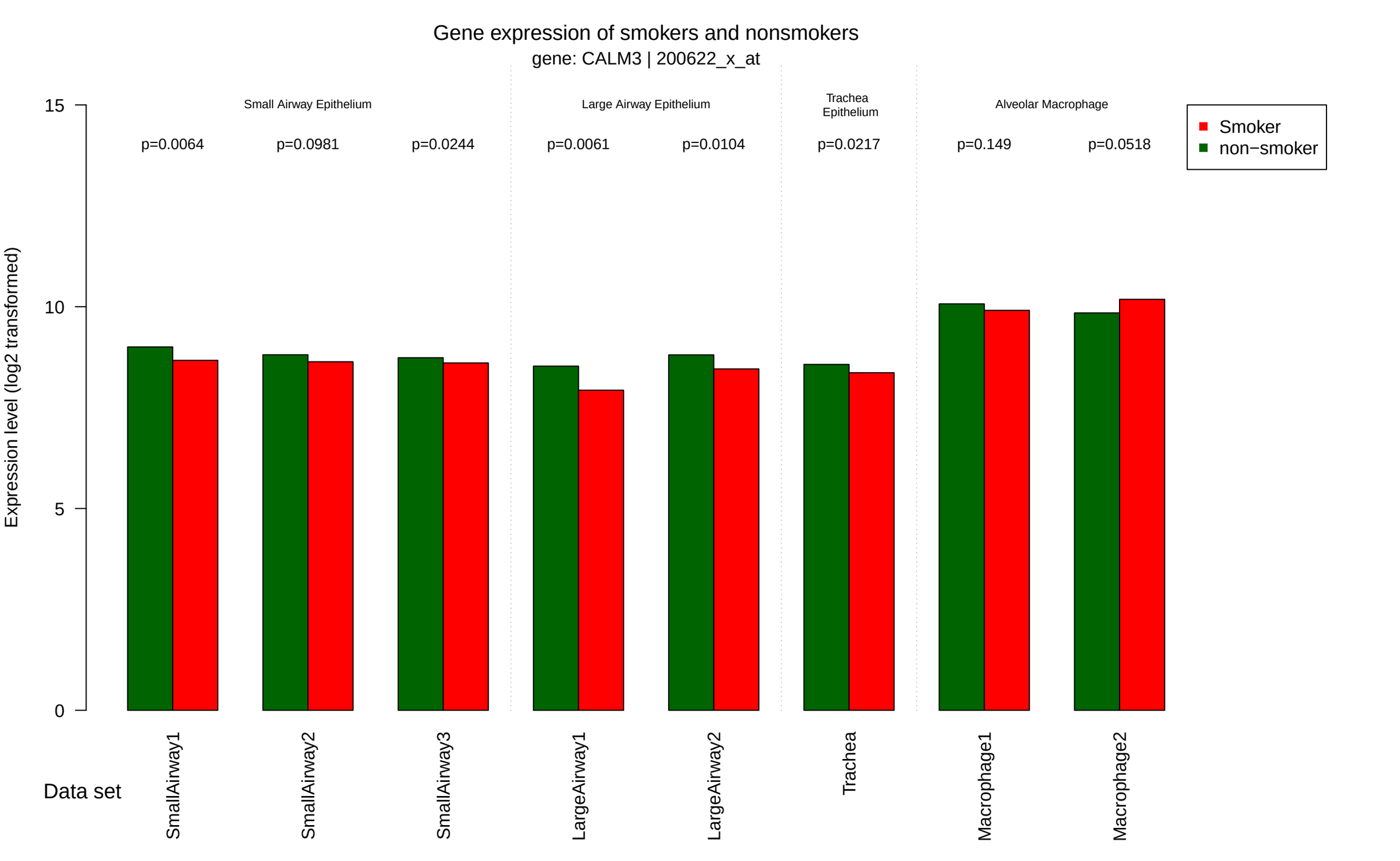
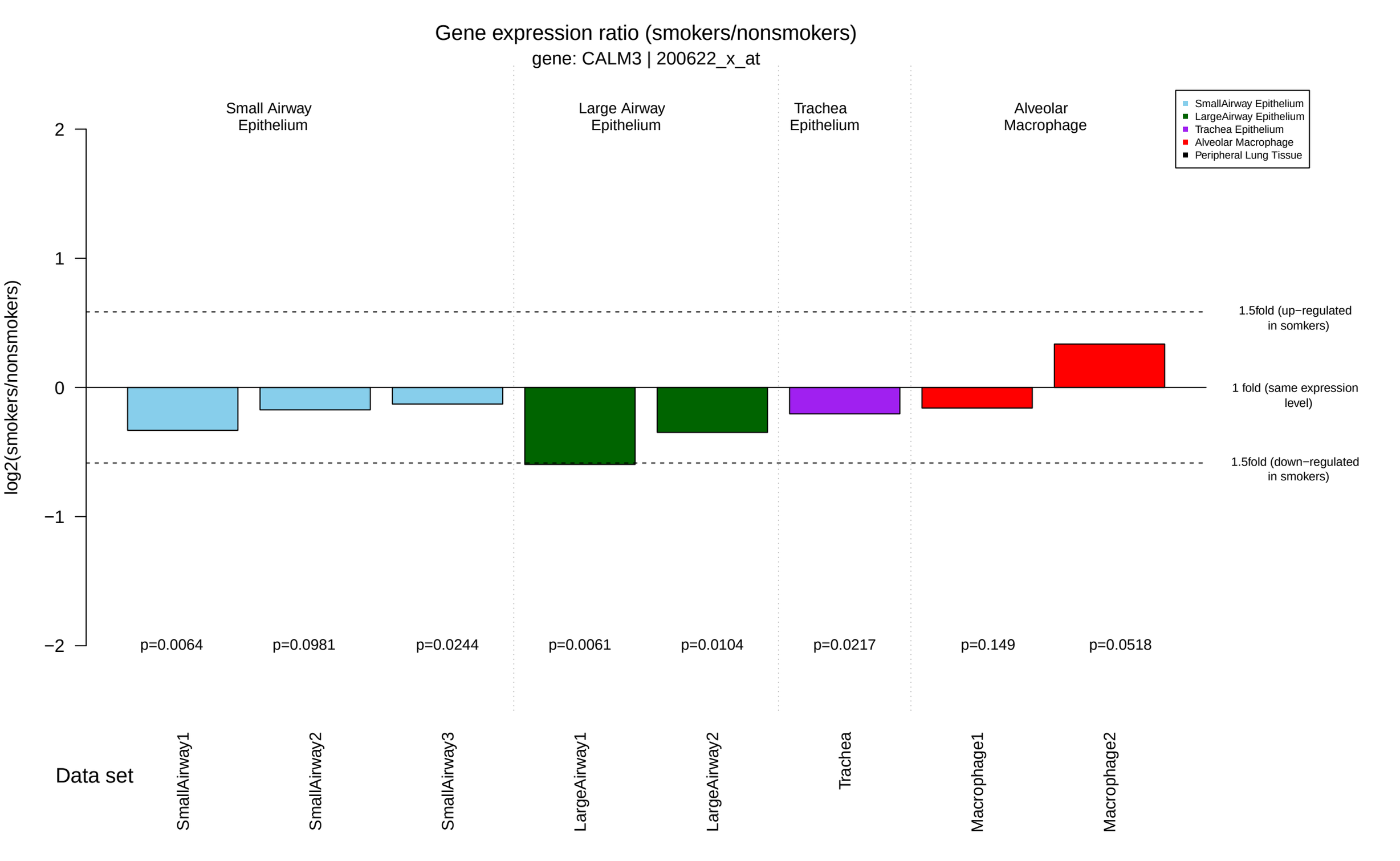
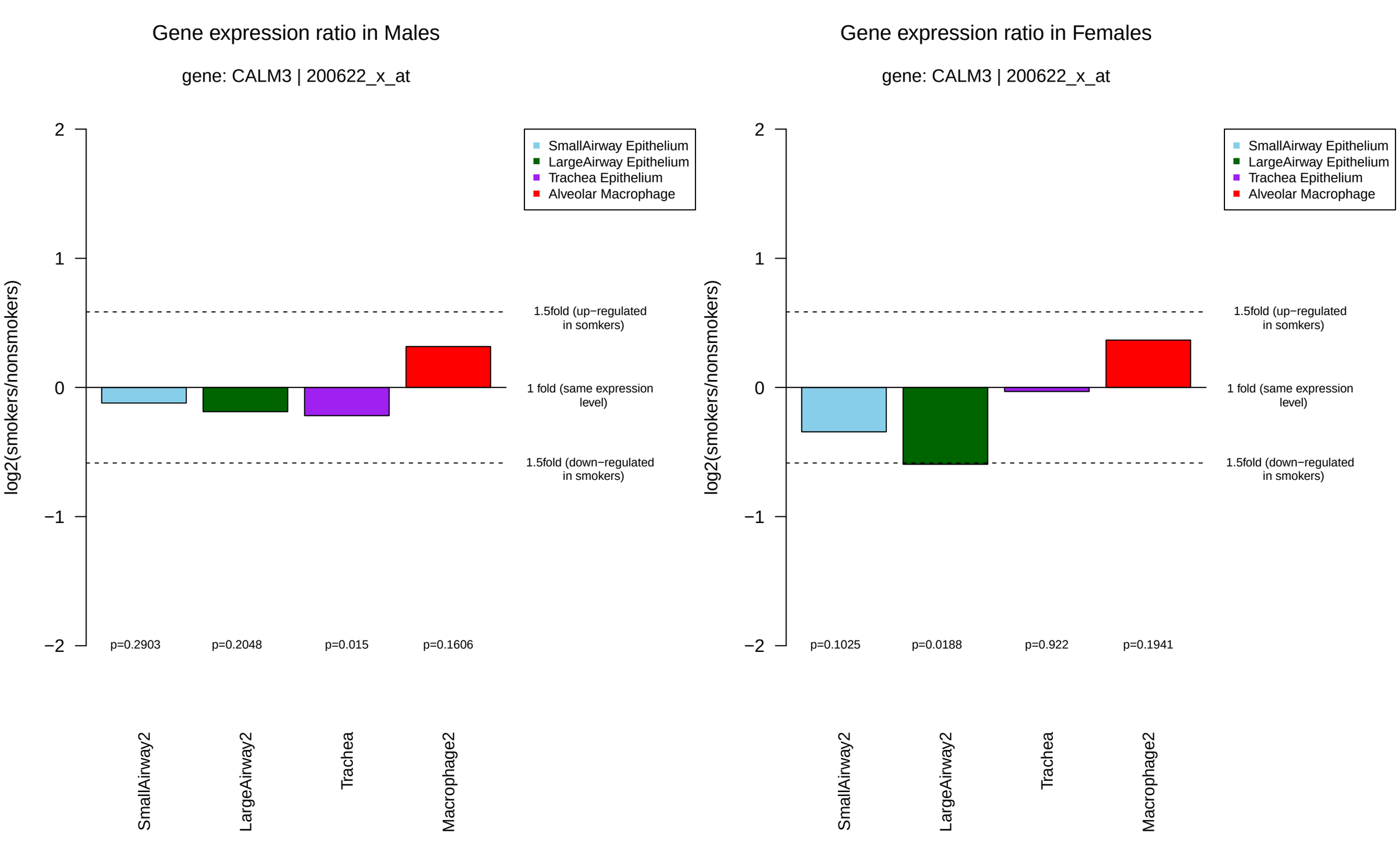
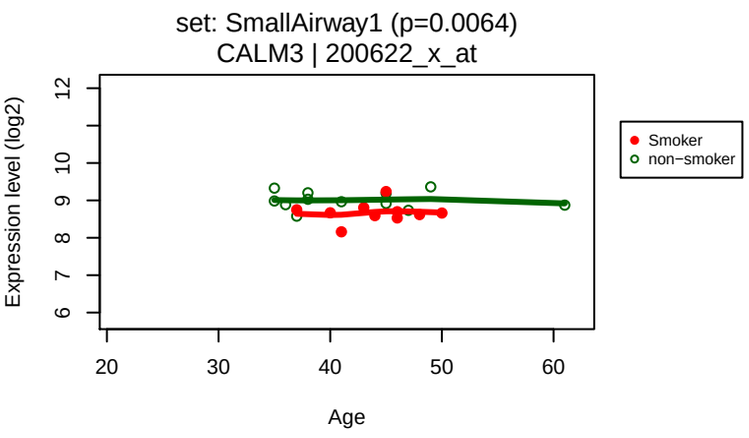
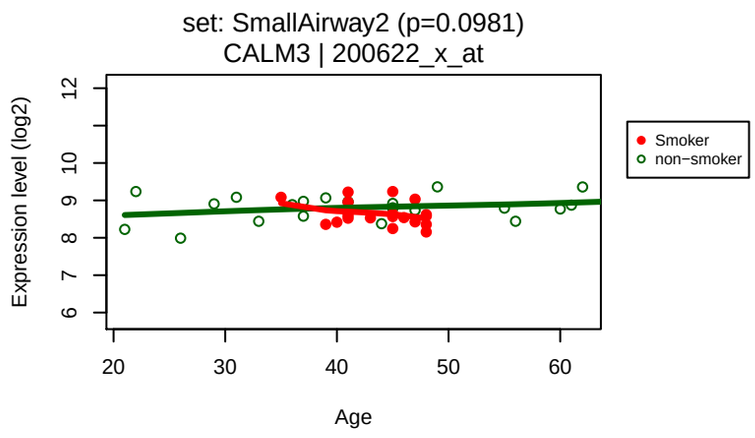
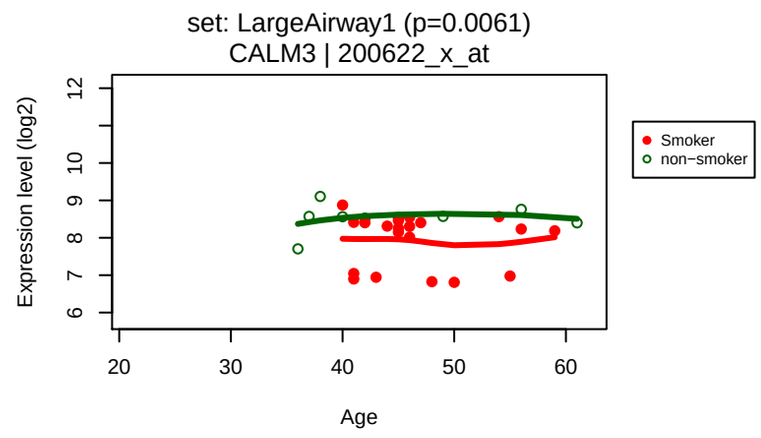

Gene expression of smokers and nonsmokers
gene: CALM3 | 200622_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0064
p=0.0981
p=0.0244
p=0.0061
p=0.0104
p=0.0217
p=0.149
p=0.0518
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

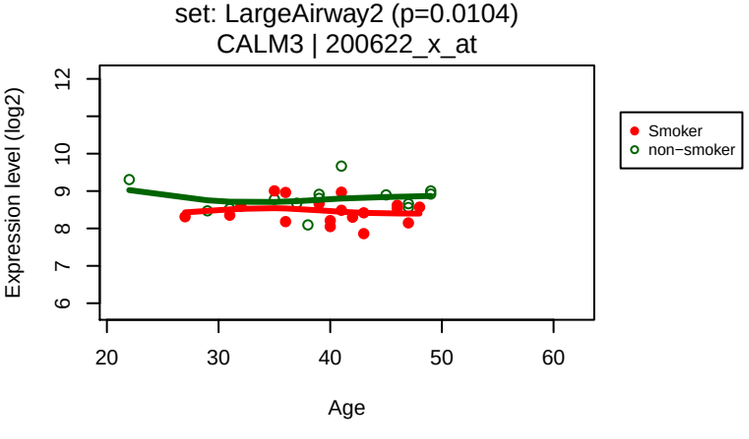

Gene expression ratio (smokers/nonsmokers)
gene: CALM3 | 200622_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0064
p=0.0981
p=0.0244
p=0.0061
p=0.0104
p=0.0217
p=0.149
p=0.0518
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

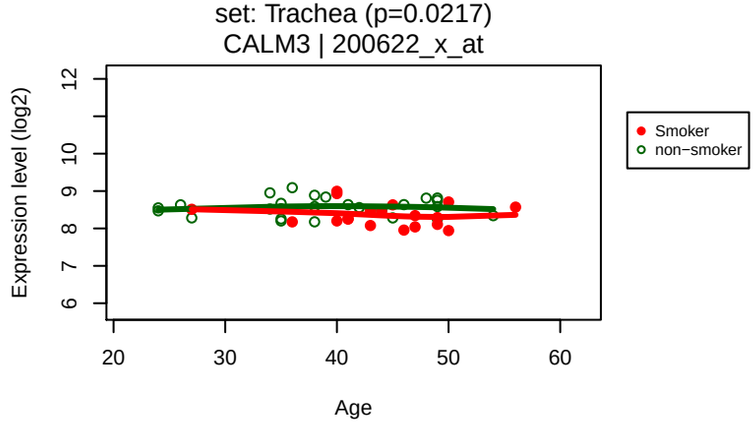

Gene expression ratio in Males
gene: CALM3 | 200622_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2903
p=0.2048
p=0.015
p=0.1606
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CALM3 | 200622_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1025
p=0.0188
p=0.922
p=0.1941
SmallAirway2
LargeAirway2
Trachea
Macrophage2

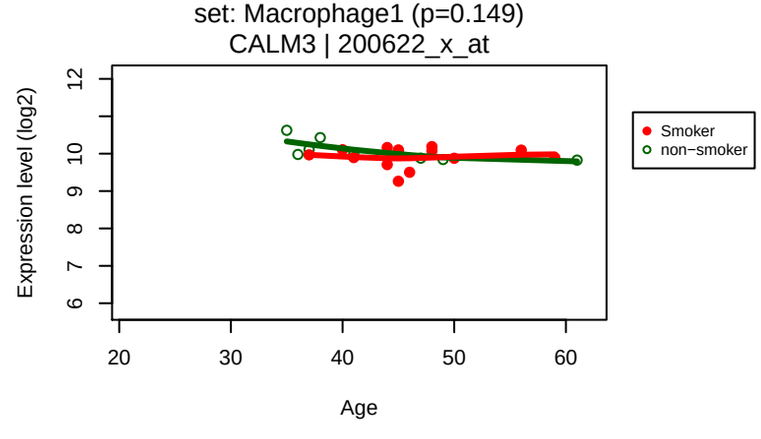

set: SmallAirway1 (p=0.0064)
CALM3 | 200622_x_at
Expression level (log2)
Age
Smoker
non−smoker

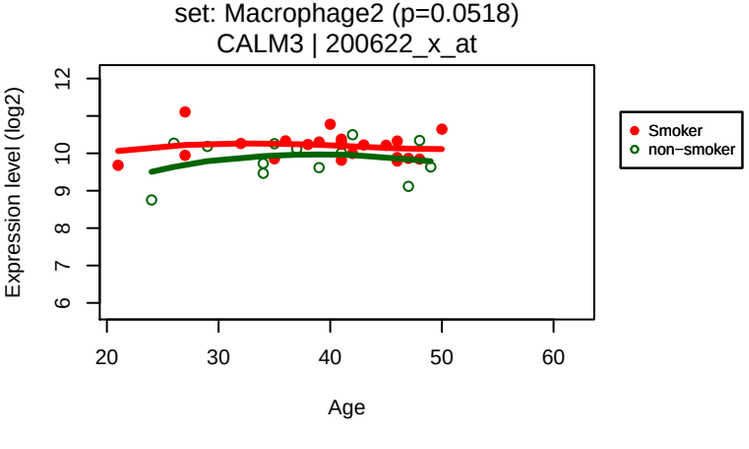

set: SmallAirway2 (p=0.0981)
CALM3 | 200622_x_at
Expression level (log2)
Age
Smoker
non−smoker

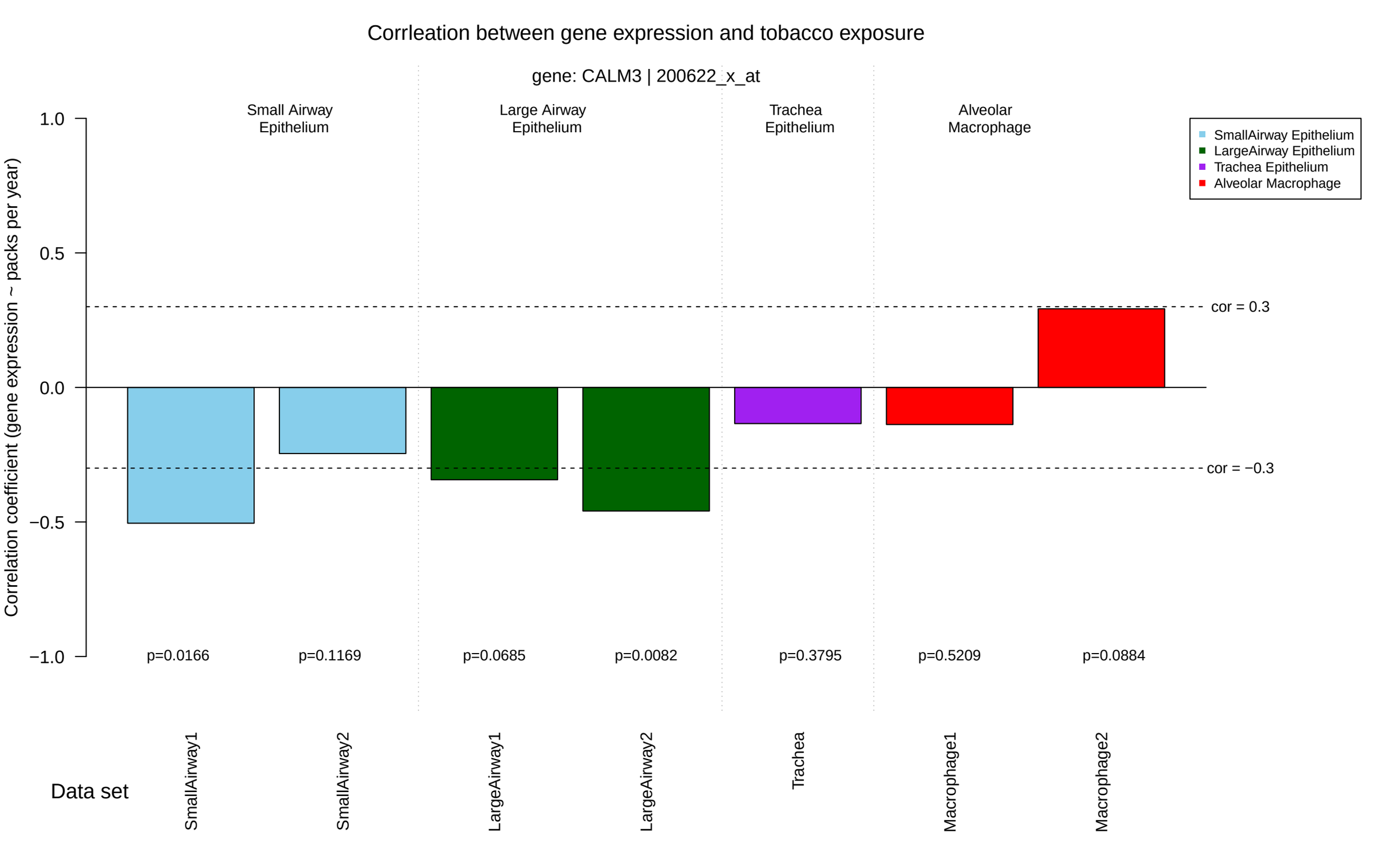

set: LargeAirway1 (p=0.0061)
CALM3 | 200622_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0104)
CALM3 | 200622_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0217)
CALM3 | 200622_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.149)
CALM3 | 200622_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0518)
CALM3 | 200622_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CALM3 | 200622_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0166
p=0.1169
p=0.0685
p=0.0082
p=0.3795
p=0.5209
p=0.0884
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2